CHAPTER 07

Integer Programming

Decision Making / Game Theory / Systems Analysis

TEXTBOOK	CHAPTER	WEEK
Modern Management Science, 9th ed.	**Ch. 7**	**Week 9**

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Made by Kitae Kim, formatted by Codex.

Overview

Introduction to Integer Programming (IP)

How to solve IP

Applications of IP

Recap: Linear Programming Assumptions

- Certainty
- Linearity
- Proportionality
- Additivity
- Divisibility: Variables may take fractional values

Integer Programming Models

- Use Integer Programming when variables **must take integer values** rather than fractions or decimals (the divisibility assumption does not hold).
- Variables represent the number of units produced
- Model either-or decisions with binary variables (0 or 1)

Example 7-1: Problem and Data

Hankook Supermarket has secured KRW 15.5 billion to expand by opening stores and storage facilities. Each new store costs KRW 1 billion and earns KRW 30 million per month. Each storage facility costs KRW 3 billion and earns KRW 72 million per month. Management plans to open at least one storage facility and no more than five stores.

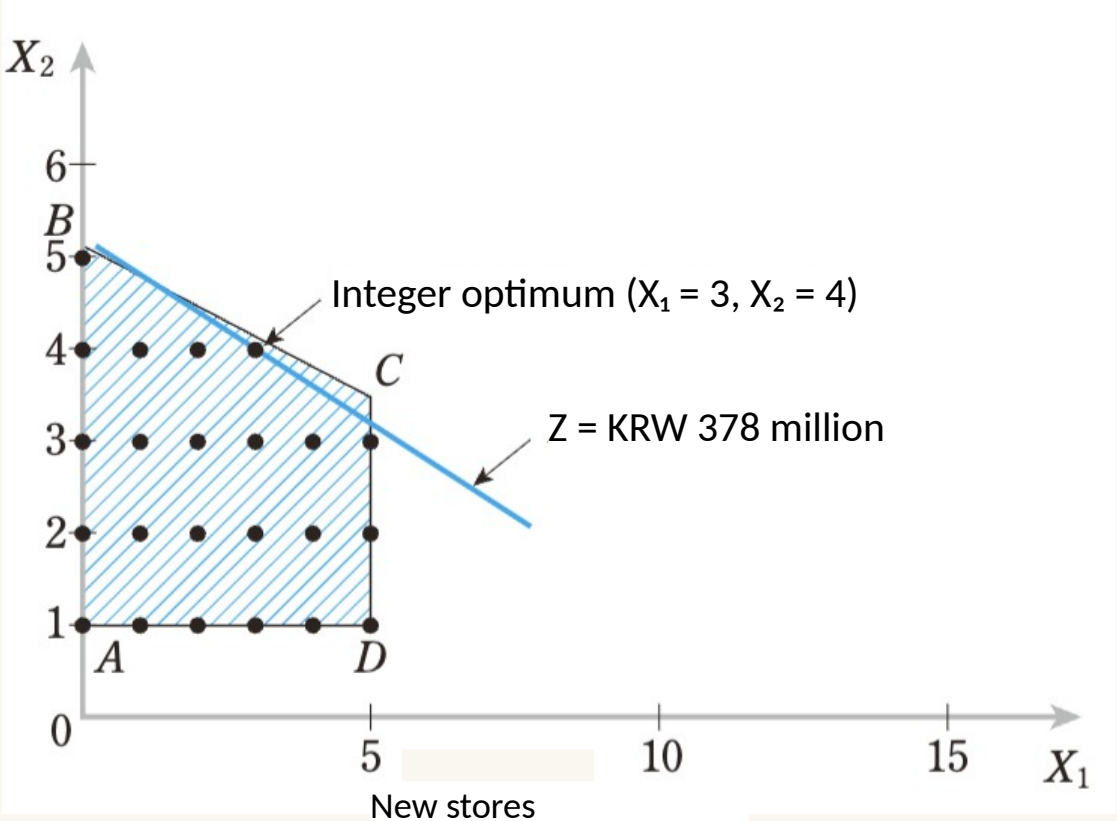
Formulate an Integer Programming model to maximize profit within the available budget.

Example 7-1: Integer Programming Model

- X1: Number of new stores
- X2: Number of new storage facilities
- Maximize Z = 0.3 X1 + 0.72 X2 (Monthly profit, unit: KRW 100 million)
- Subject to: 10X1 + 30X2 <= 155 (Budget, unit: KRW 100 million)
  X1 <= 5 (Store limit)
  X2 >= 1 (Minimum storage facilities)
  X1 , X2 >= 0
  X1, X2 are integers

Example 7-1: Integer Solution

Solution to the Integer Programming model



Introduction to Integer Programming (IP)

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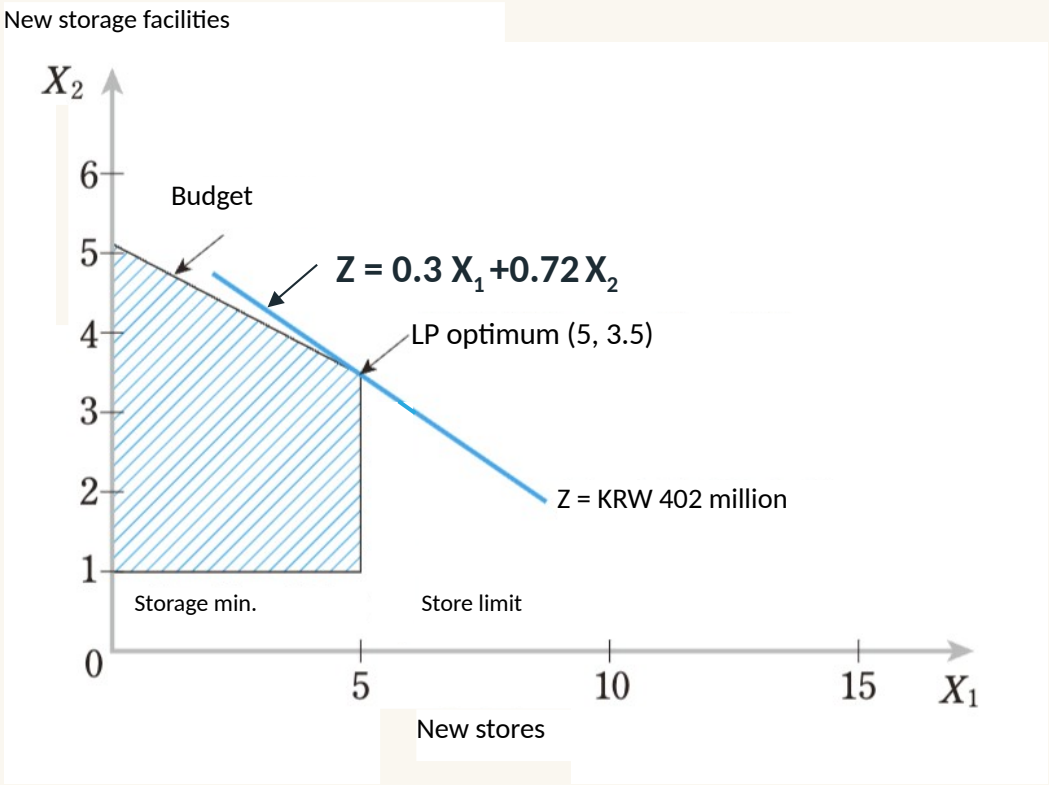
How to solve IP models?

- LP relaxation
- Branch-and-bound method

Linear Programming Relaxation

- LP relaxation: Ignore integrality constraints and find the optimal solution
- Then adjust any fractional values to integers

- LP relaxation optimum:
- Is (5, 3) or (5, 4) optimal?
- What about (3, 4)?

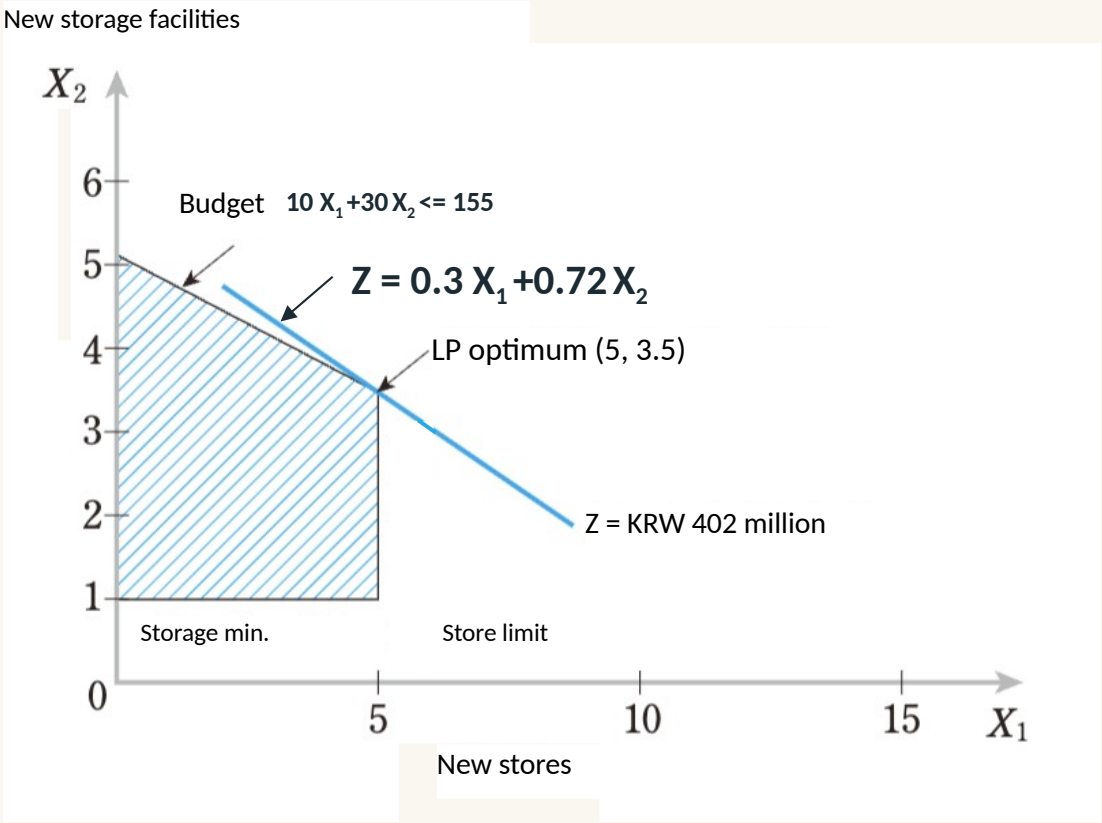


Branch-and-Bound Method

- Branch-and-bound divides the feasible region into subproblems and analyzes each separately.
- An efficient method for solving Integer Programming models
- For Example 7-1, the LP relaxation optimum is (X1 = 5, X2 = 3.5).
- Branch into X2 <= 3 and X2 >= 4, then solve each subproblem.

Branch-and-Bound: First Branch

Case: $X_2 \leq 3$

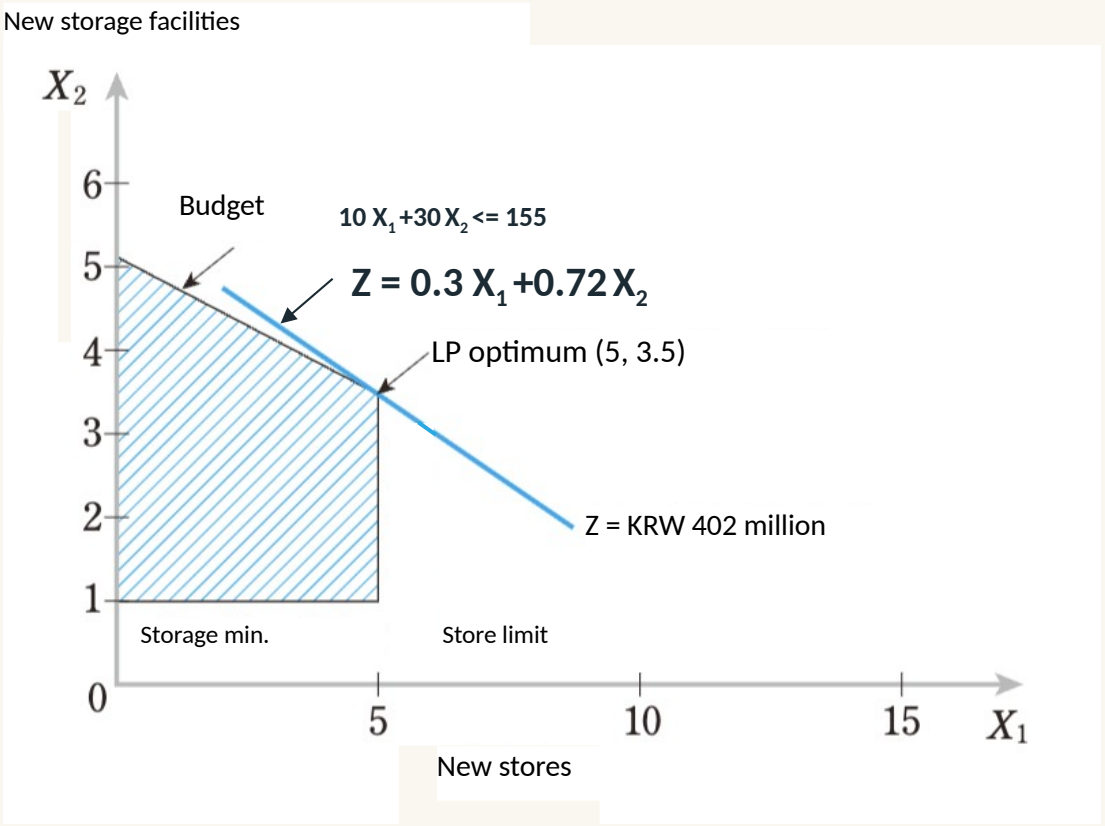


Branch-and-Bound: Further Branching

Case: $X_2 \geq 4$

$X_1 \geq 4$

$X_1 \leq 3$



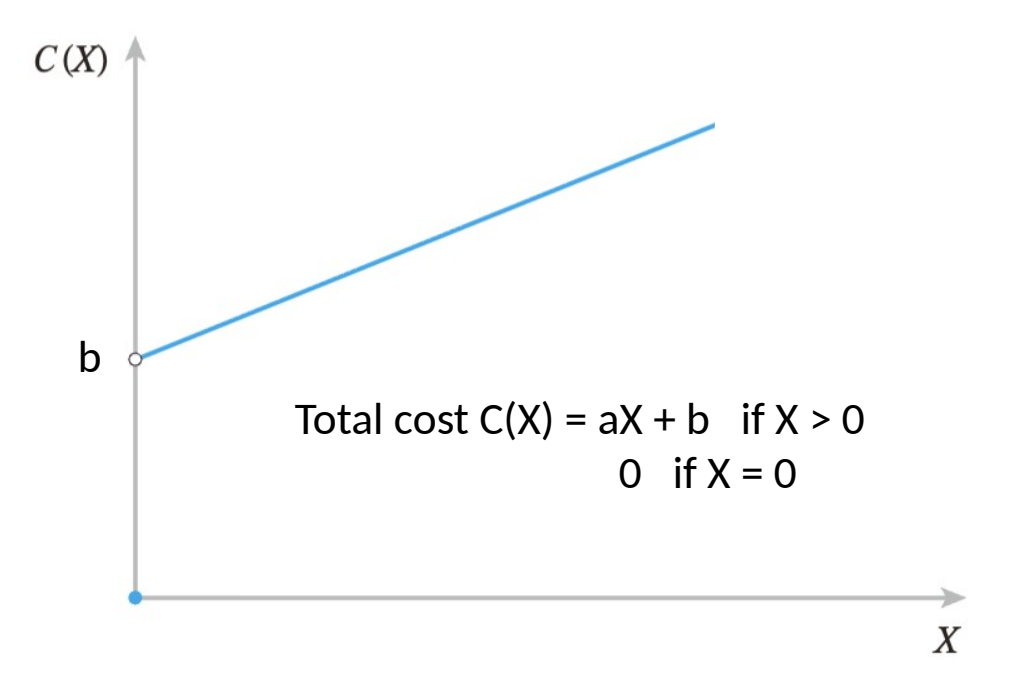
Introduction to Integer Programming (IP)

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Fixed-Cost Problems

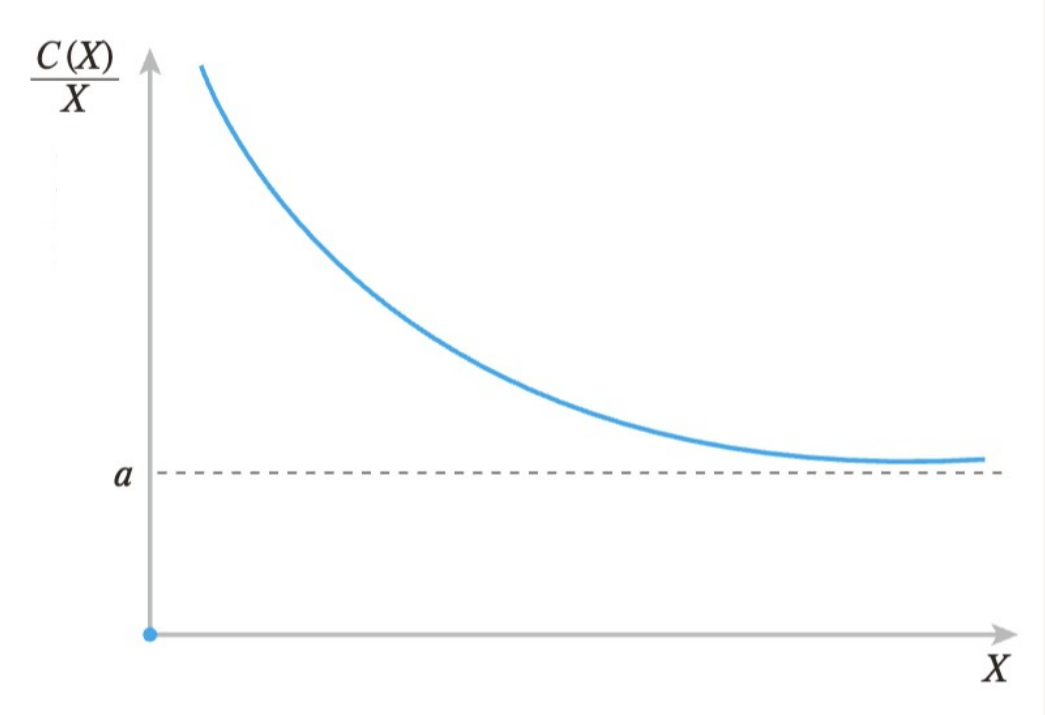
- Linear programming assumes variable costs only, **with no fixed costs**
- In practice, firms also incur fixed costs
- Use Integer Programming to model fixed costs
- This brings us to mixed integer programming



Fixed Costs and Economies of Scale

- Fixed costs are a common source of economies of scale.
- Unit cost C(X)/X decreases as output increases.
- Add a binary variable Y and link it to X:
  - Y = 1 if X > 0 (producing)
  - 0 if X = 0 (not producing)
- X ≤ cY, where c is production capacity

<Economies of Scale>



Example 7-3: Problem and Data

K Manufacturing must produce 20,000 units using three machines. Each machine incurs a fixed cost when activated. The table lists each machine's fixed cost, variable cost, and production capacity. Formulate an Integer Programming model to choose which machines to use and how much each should produce to minimize total production cost.

Machine	Fixed cost (KRW 10,000)	Variable cost (KRW 10,000/unit)	Capacity (units)
1	300	2	8,000
2	100	10	13,000
3	200	5	14,000

Example 7-3: Model and Solution

Y_j = 1 if machine j operates
0 otherwise
X_j = Output of machine j, j = 1, 2, 3

$$\begin{aligned}
\text{minimize} \quad & 300Y_1 + 100Y_2 + 200Y_3 + 2X_1 + 10X_2 + 5X_3 \\
\text{subject to} \quad & X_1 + X_2 + X_3 \geq 20{,}000 \\
& X_1 \leq 8{,}000Y_1 \\
& X_2 \leq 13{,}000Y_2 \qquad X_1, X_2, X_3 \geq 0 \\
& X_3 \leq 14{,}000Y_3 \qquad Y_1, Y_2, Y_3 \in \{0, 1\}
\end{aligned}$$

Optimal solution from CVXPY

Total cost: KRW 765 million

Y1 = 1

Y2 = 0

Y3 = 1

X1 = 8,000

X2 = 0

X3 = 12,000

Integer Programming with CVXPY

- Many real-world problems use mixed-Integer Programming models
- These models optimize over both integer and continuous variables
- CVXPY can solve integer and mixed-integer models using a solver that supports mixed-integer programming (MIP).

Problem type

Solver	LP	QP	SOCP	SDP	EXP	POW	MIP
CBC	X						X
CLARABEL	X	X	X	X	X	X	
COPT	X	X	X	X	X		X**
DAQP	X	X					
GLOP	X						
GLPK	X						
GLPK_MI	X						X
OSQP	X	X					
PIQP	X	X					
PROXQP	X	X					
QPALM	X	X					
PDLP	X						

Integer Programming with CVXPY

- If x must be integer:
  - x = cp.Variable( )
- If y must be 1 or 0:
  - y = cp.Variable( )

Example 7-2: Investment Selection

H Group is considering three projects over the next three years. The table gives each project's annual funding needs and present value of profit.

Table 7.1 Investment plan (KRW 100 million)

Project	Profit PV	Year 1	Year 2	Year 3
Factory expansion	450	100	100	100
New warehouse	200	150	50	50
Advanced machinery	350	50	200	100
Available funds		250	400	200

Present value discounts future investments and returns to today, then sums them. At a 10% annual rate, a project invests KRW 1 billion now and KRW 1 billion in year 1, then earns KRW 300 million annually from year 2 onward.

PV = −10 − 10/1.1 + 3/1.1² + 3/1.1³ + …
= −10 − 10/1.1 + (3/1.1²)[1/(1 − 1/1.1)]
≈ 8.2 (KRW 100 million)

$$
\begin{aligned}
\text{Maximize} \quad & 450F + 200W + 350M \\
\text{subject to} \quad & 100F + 150W + 50M \le 250 \\
& 100F + 50W + 200M \le 400 \\
& 100F + 50W + 100M \le 200 \\
& F, W, M \text{은 } 0 \text{ 또는 } 1.
\end{aligned}
$$

F, W, M ∈ {0, 1}

Example 7-2: CVXPY Model and Solution

$$
\begin{aligned}
\text{Maximize} \quad & 450F + 200W + 350M \\
\text{subject to} \quad & 100F + 150W + 50M \leq 250 \\
& 100F + 50W + 200M \leq 400 \\
& 100F + 50W + 100M \leq 200 \\
& F, W, M \text{은 } 0 \text{ 또는 } 1.
\end{aligned}
$$

F, W, M ∈ {0, 1}

Status: optimal · Profit PV: 800.0
Factory: 1 · Warehouse: 0 · Machinery: 1

```
import cvxpy as cp

# 1. Binary project decisions
X1 = cp.Variable(boolean=True) # Factory
X2 = cp.Variable(boolean=True) # Warehouse
X3 = cp.Variable(boolean=True) # Machinery

# 2. Maximize present value of profit
objective = cp.Maximize(
    450*X1 + 200*X2 + 350*X3)

# 3. Annual funding constraints
constraints = [
    100*X1 + 150*X2 + 50*X3 <= 250,
    100*X1 + 50*X2 + 200*X3 <= 400,
    100*X1 + 50*X2 + 100*X3 <= 200
]

# 4. Define and solve the problem
problem = cp.Problem(objective, constraints)
problem.solve()

# 5. Results (profit in KRW 100 million)
print("Status:", problem.status)
print("Profit PV:", problem.value)
print("Factory:", round(float(X1.value)))
print("Warehouse:", round(float(X2.value)))
print("Machinery:", round(float(X3.value)))
```

Summary

- **Introduction to Integer programming (IP)**
  - Use when decision variables must be integers
  - LP divisibility assumption does not hold
  - Binary variables (0/1)
- **How to solve IP**
  - LP relaxation
  - Branch-and-bound:
- **Applications of IP**
  - Mixed-integer programming (MIP): Fixed costs
  - Use a MIP-capable solver in CVXPY to solve integer and mixed-integer models